Coefficient of correlation r= 0.9. Estimate the likely sales for a proposed advertisement expenditure of Rs. 10 crores.

Solution:

Let the sales be X and advertisement expenditure be Y

Given $\overline{X}$ =40, $\overline{Y}$ =6, σ_x=10, σ_y=1.5 and r=0.9

Equation of line of regression x on y is

$$X-\overline{X} = r\frac{\sigma_x}{\sigma_y}(Y-\overline{Y})$$

$$X-40 = (0.9)\frac{10}{1.5}(Y-6)$$

$$X-40 = 6Y-36$$

$$X = 6Y+4$$

When advertisement expenditure is 10 crores i.e., Y=10 then sales X=6(10)+4=64 which implies sales is 64.

Example 9.15

There are two series of index numbers P for price index and S for stock of the commodity. The mean and standard deviation of P are 100 and 8 and of S are 103 and 4 respectively. The correlation coefficient between the two series is 0.4. With these data obtain the regression lines of P on S and S on P.

Solution:

Let us consider X for price P and Y for stock S. Then the mean and SD for P is considered as $\overline{X}$ =100 and σ_x=8. respectively and the mean and SD of S is considered as $\overline{Y}$ =103 and σ_y = 4. The correlation coefficient between the series is $r(X,Y)$ = 0.4

Let the regression line X on Y be

$$X-\overline{X} = r\frac{\sigma_x}{\sigma_y}(Y-\overline{Y})$$

$$X-100 = (0.4)\frac{8}{4}(Y-103)$$

$$X-100 = 0.8(Y-103)$$

X–0.8Y–17.6 = 0 (or) X=0.8Y+17.6

The regression line Y on X be

$$Y-\overline{Y} = r\frac{\sigma_y}{\sigma_x}(X-\overline{X})$$

$$Y-103 = (0.4)\frac{4}{8}(X-100)$$

$$Y-103 = 0.2\,(X-100)$$

$$Y-103 = 0.2\,X-20$$

$$Y = 0.2\,X+83 \text{ (or) } 0.2\,X-Y+83=0$$

Example 9.16

For 5 pairs of observations the following results are obtained ΣX=15, ΣY=25, ΣX^2=55, ΣY^2=135, ΣXY=83 Find the equation of the lines of regression and estimate the value of X on the first line when Y=12 and value of Y on the second line if X=8.

Solution:

Here N=5, $\overline{X}=\frac{\Sigma X}{N}=\frac{15}{5}=3$, $\overline{Y}=\frac{\Sigma Y}{N}=\frac{25}{5}=5$ and the regression coefficient

$$b_{xy}=\frac{N\Sigma XY-\Sigma X\Sigma Y}{N\Sigma Y^2-(\Sigma Y)^2}=\frac{5(83)-(15)(25)}{5(135)-(25)^2}=0.8$$

The regression line of X on Y is

$$X-\overline{X} = b_{xy}(Y-\overline{Y})$$

$$X-3 = 0.8(Y-5)$$

$$X = 0.8Y-1$$

When Y = 12, the value of X is estimated as

$$X = 0.8\,(12)-1=8.6$$

The regression coefficient

$$b_{yx} = \frac{N\Sigma XY-\Sigma X\Sigma Y}{N\Sigma X^2-(\Sigma X)^2} = \frac{5(83)-(15)(25)}{5(55)-(15)^2}=0.8$$

Thus b_{yx}= 0.8 then the regression line Y on X is

$$Y-\overline{Y} = b_{yx}(X-\overline{X})$$

$$Y-5 = 0.8(X-3)$$

$$Y = 0.8X+2.6$$

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